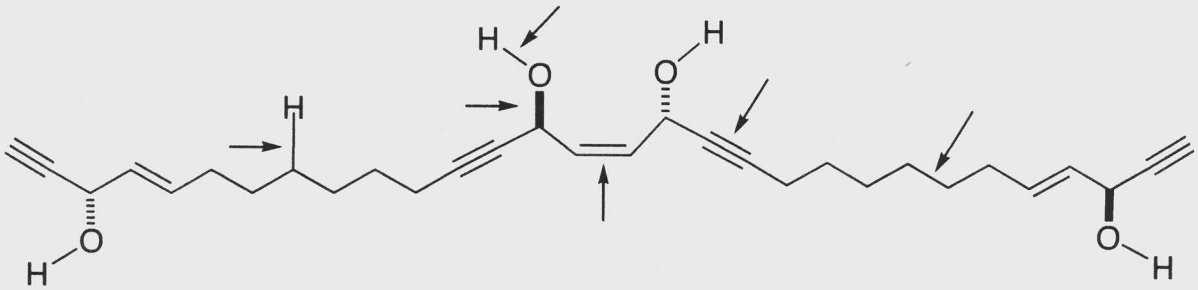
Organic Division Cumulative Exam September 9, 2011

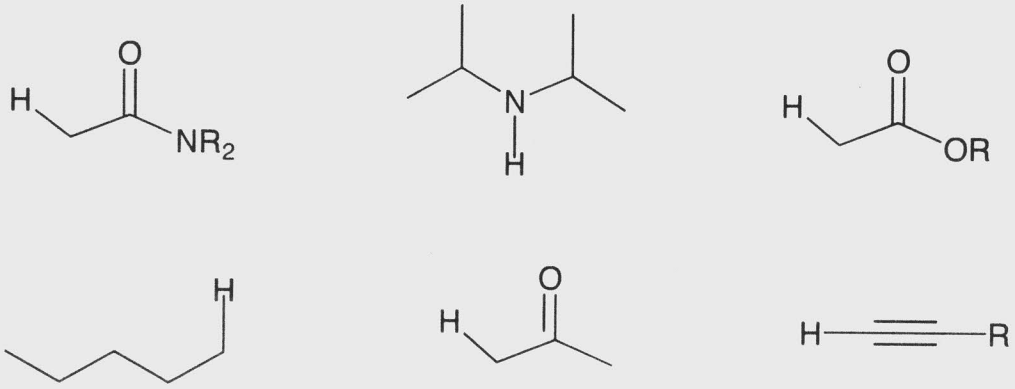
Topic: Important numbers in organic chemistry. There are not a lot of numbers you need to know in organic chemistry, but this cume will test your knowledge some of the more important ones, including bond lengths, strengths, pK_a's and A-values. **If you don't know these numbers, you should familiarize yourself with them.**

1. Give the bond lengths to within 0.1 Å for an average bond of the indicated type.
2. Give the bond strengths (homolytic bond dissociation energies) rounded to the nearest 10 kcal/mol. For the multiple bonds, give the **total** bond energy (i.e. single + double, or single + double + triple).



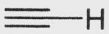
A cytotoxic polyacetylene isolated from the marine sponge *Petrosia*

3. Rank the acidity from most to least acidic and give the approximate pK_a's of the following compounds.



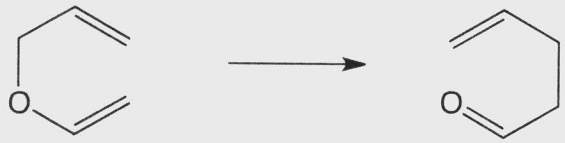
0-1

4. Give approximate A-values in kcal/mol (A-value = free energy change for the conversion of the equatorial to axial substituent in cyclohexane).

R	A value
	
$C(CH_3)_3$	
OCH_3	
CH_3	
phenyl	
CH_2CH_3	
$CH{=}CH_2$	

5. Estimate ΔH for the following reactions. Show how you arrived at your answers.

(a)



(b)

0-2